THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 4: **Exploring Magnets** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The reason a magnet attracts iron strongly at the ends is that: 1

(a) the ends are heavy **(b)** the ends are sharp
(c) the ends are coloured **(d)** the force is concentrated at the poles

2. Which of these is a way to magnetise a steel needle? 1

(a) hit it **(b)** stroke it with a magnet in one direction
(c) heat it **(d)** paint it

3. Why does a magnetic compass needle sometimes point wrongly near a steel almirah? 1

(a) Earth's field is gone **(b)** wooden floor
(c) needle is broken **(d)** steel disturbs the field

4. What would a freely suspended magnet do if there were no Earth's magnetic field? 1

(a) point north **(b)** spin always
(c) stay in any direction **(d)** point south

5. A compass needle placed near a bar magnet is deflected because: 1

(a) wind blows it **(b)** air pushes it
(c) heat moves it **(d)** the magnet's field acts on it

6. Magnetic materials can be separated from non-magnetic materials by: 1

(a) magnetic separation **(b)** sieving
(c) evaporation **(d)** filtration

7. Which of the following is not a magnetic material? 1

(a) iron **(b)** cobalt **(c)** brass **(d)** nickel

8. Why do we use steel in making permanent magnets? 1

(a) it is soft **(b)** it is cheap
(c) it holds magnetism for long **(d)** it is transparent

9. Which pair of items can both be picked up by a magnet? 1

(a) iron nail and plastic cap **(b)** pin and cork
(c) paper and rubber **(d)** iron nail and steel pin

10. A magnet is made of: 1

(a) cotton **(b)** glass **(c)** paper **(d)** steel or alloy

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why can't aluminium foil be picked with a magnet? 2

12. Describe why a freely suspended magnet is used as a compass. 2

13. Why is a keeper placed on magnets? 2

14. How do you make a magnet lose strength deliberately? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Draw (by words) the pattern of iron filings around a bar magnet. 3

16. How are magnets used in the speaker? 3

17. Explain the 'north-seeking' term. 3

OR

A magnet loses its strength after being heated. Explain. 3

18. Why is a freely hanging magnet useful for finding the direction in a ship? 3

OR

How does a refrigerator magnet keep the door closed? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe how magnets are used for separating magnetic from non-magnetic waste in a recycling plant. 5

OR

Describe how a magnetic compass is used to find direction in a forest. 5

20. Explain the significance of the north-south property of a freely suspended magnet and how it led to the compass. 5

OR

How would you design a test to find which of three unlabelled bars is a magnet, iron and wood? 5

END OF QUESTION PAPER — Total: 40 Marks

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